6.2 Output Signals

6.2.1 Alphabetical Overview

Signal name	Page
CONTROL READY	6–19
SHEET THICKNESS	6–17
SHEET THICKNESS TOLERANCE	6–17
WELD COMPLETE (WC)	6–18
TIPDRESSING REQUEST x	6–22
TIP REPLACEMENT x	6–16
TIPDRESS PREWARNING x	6–22
END OF STEPPER ELECTRODE x	6–23
WELD/NO WELD	6–21
WITHOUT REGULATION AND WITHOUT MONITORING FUNCTION	6–22
WITHOUT MONITORING FUNCTION	6–21
AUTOMATIC PROGRAM SELECTION ACTIVE	6–22
PROP. VALVE SELECTION	6–16
SPOT SELECTION TAKEN OVER	6–23
WELDING FAULTS	6–20
STATUS	6–24
TIPDRESS PREWARNING x	6–23
PREWARNING TIP REPLACEMENT x	6–15

6.2.2 Prewarning Tip Replacement x

Informs the PLC/the robot that the tip in question (1, 2, 3, or 4) requires replacement as soon as possible.
The signal is set when the respective parameterizable tip prewarning value has been reached.
The ”Tips have been replaced” acknowledgement (see page 6–13) resets ”Prewarning tip replacement”.

☞ **For more information, refer to ”Tip Replacement” on page 3–29.**

I/O Signal Descriptions

6.2.3 Tip Replacement x

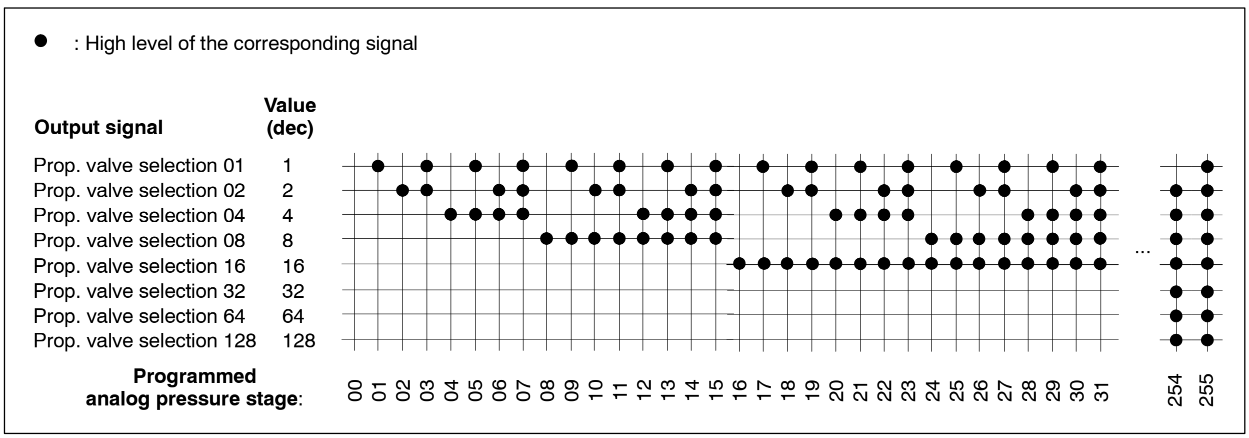
Informs the PLC/the robot that the cutters of the tip in question (1, 2, 3, or 4) require replacement as soon as possible.
The signal is set when the parameterizable value ”Max. tip wear” Has been reached; it can be reset by means of the programming device (BOS) or via the ”Tip replacement” acknowledgment (see page 6–13).

☞ **For more information, refer to ”Tip Replacement” on page 3–29.**

6.2.4 Prop. Valve Selection

By means of the 8 output signals ”Prop. Valve selection 01” to ”Prop. Valve selection 128”, the timer transmits the programmed manipulated power variable via DeviceNet.

To this end, the maximum possible manipulating range at X2 (see page 5–7) is subdivided into 256 discrete steps, and the active output signal of X2 is output as binary–coded pressure stage.



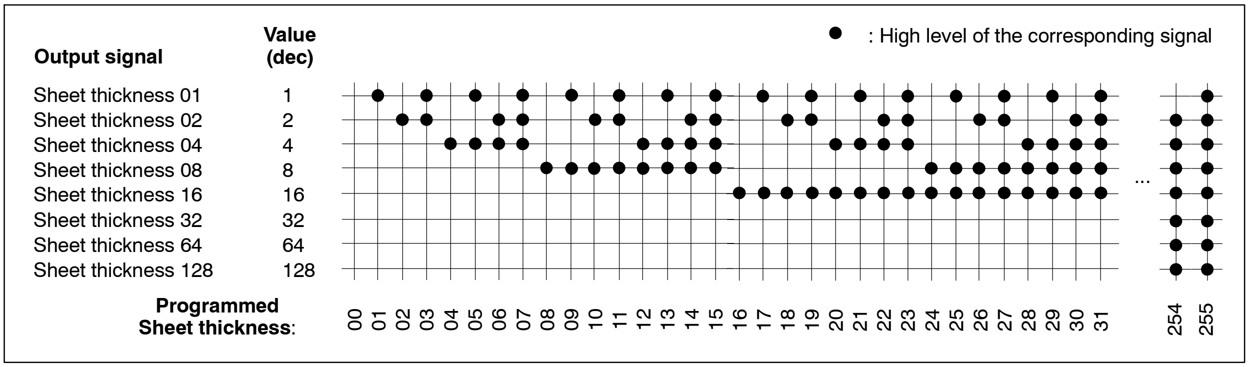
Binary–coded output of the active pressure stage

☞ **The signals are used in connection with the ”Sheet thickness” and ”Sheet thickness tolerance” signals to active servotongs.**

I/O Signal Descriptions

6.2.5 Sheet Thickness

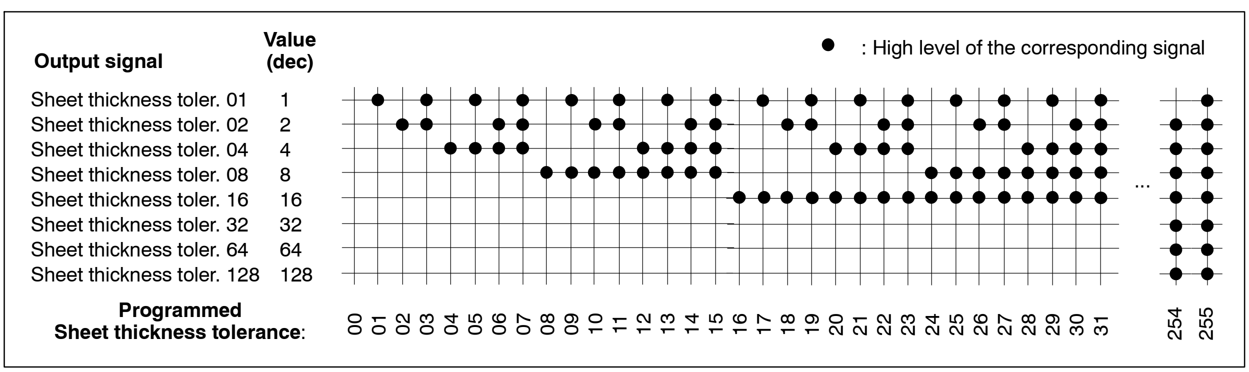
Via the 8 output signals ”Sheet thickness 01” to ”Sheet thickness 128”, the timer transmits the programmed sheet thickness in binary code via DeviceNet to the electronic tongs system. In this way, the closing motion of the tongs in a short distance of the material surface can be influenced in such a way that the electrodes make contact ”softly”.



Binary–coded output of sheet thickness

6.2.6 Sheet Thickness Tolerance

Via the 8 output signals ”Sheet thickness tolerance 01” to ”Sheet thickness tolerance 128”, the timer transmits the programmed sheet thickness tolerance in binary code via DeviceNet to the electronic tongs system. In connection with the transmitted sheet thickness (see Chap. 6.2.5), the closing motion of the tongs in a short distance of the material surface can be influenced in such a way that the electrodes make contact ”softly”.



Binary–coded output of sheet thickness tolerance

6.2.7 Weld Complete (WC)

The output signal ”Weld complete” informs the connected peripherals (PLC/robot) about the termination of the welding process.
This way, the next step of the working process can be initiated. The logics for generating a WC is activated in the following cases:

1. in case of single spot welding (e.g. in connection with robots) after each spot
2. in case of seam mode (e.g. roll seam welds) at the end of the seam
3. upon ”set WC” (possible only via BOS)
4. upon ”Reset Fault with WC” (see page 6–7).

☞ **How long the WC remains set, depends on the input signal ”Start”. See ”WC period”.**

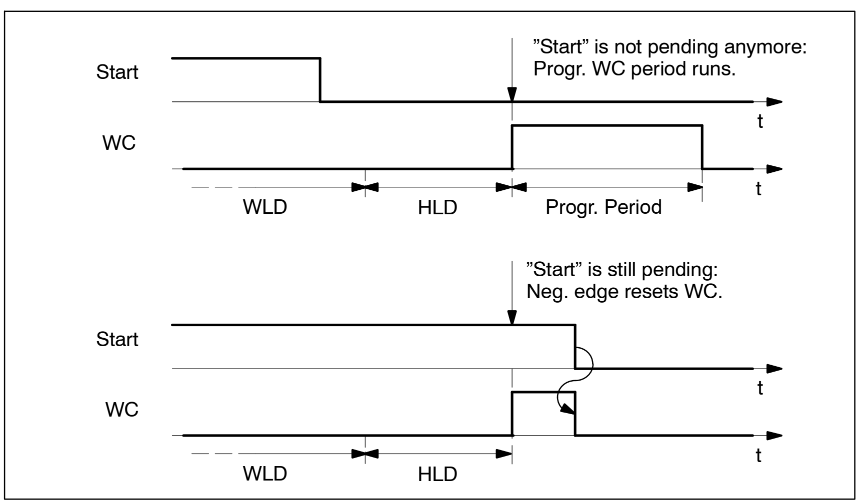
The 1. and 2. case give you the possibility to adjust the WC to your application via parameterization (BOS).

- Automatic output of the WC only after a proper weld, or even after an erroneous weld.
- Time at which the WC should be set (see ”WC starting time”).

WC period

Normally, the timer resets the signal ”Weld complete” automatically if it recognizes a negative edge at the input ”Start”.
Nevertheless, there are situations possible in which the ”Start” signal is reset prior to setting the WC. Here, a triggering on the negative edge of ”Start” is not possible. Therefore, when setting the WC, the timer checks whether ”Start” is still pending and reacts as follows:

”Start” is set:	WC will be reset only after a negative edge of ”Start”
Start” is not set:	WC will be reset after the parameterized WC period (GUI (BOS); default: 20 ms).



WC period depends on signal ”Start”

☞ **The programmed WC period also runs during start simulation (BOS)!**

WC starting period

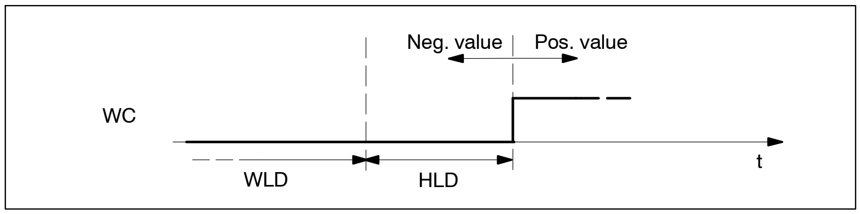
The time, at which "Weld complete" shall be output, is **– in relation to the end of the HLD –** parameterizable within the following limits:

- with PSI: +/−1000 ms
- with PST: +/−50 cycles.

Restriction: The earliest possible output of the WC is possible after 20 ms or 1 cycle after the start of the HLD.

Default setting: End of HLD minus 20 ms or 1 cycle (see "example").

This way, the starting time of the WC can be anticipated and moved into the HLD time as well as delayed.



Moving of WC starting time

Example: "Fast robot communication"

If "Weld complete" is used as a command for positioning the robot to the next spot, you can compensate constant response times (caused by signal processing in the robot, drive, or PLC area) by anticipating the starting time of the WC. This will lead to shorter clock times.

CAUTION
Damage to the installation caused by positioning movements with a closed gun are possible!
Therefore, make sure that the electrodes are already open when the robots' drives start!

6.2.8 Control Ready

The output signal indicates that the timer is ready to weld.
In this state,

- you can start a new welding schedule (see page 6−2)
- the LED READY at the timer's front is lit (see page 2−8).

If an error occurs, the timer goes into ”Block”.
In this state,

- it is not possible to start a welding process (schedule)
- the LED READY at the timer’s front goes off
- the output signal ”Control Ready” will be reset.

☞ **Error and status messages are listed in the ”Error list PS5000/PS6000” (No.: 1070 087 000).**

In order to restore the ”Ready”-state of the timer after an error, you have the following possibilities at your disposal:

1. Push the reset button at the timer’s front (see page 2–8) or
2. a positive edge of the ”Fault reset” input signal (refer to page 6–6) or
3. a positive edge at the ”Reset fault with WC” input signal (refer to page 6–7) or
4. a positive edge at the ”Reset fault with spot repeat” input signal (refer to page 6–8) or
5. operation via software (”Reset fault”, ”Reset fault with WC”, and ”Reset fault with spot repeat” are also possible via BOS).

6.2.9 Welding faults

If an error occurs during welding,

- the timer sets the output ”Welding Fault” and
- deletes the output signal ”Control Ready” (refer to page 6.2.8).

Further welding processes can only be started, when all pending errors are eliminated and acknowledged (reset). Please refer to ”Reset fault” on page 6–6.

☞ **Error and status messages are listed in the ”Error list PS5000/PS6000” (No.: 1070 087 000).**

☞ **It depends on the parameterization of the timer (BOS; fault allocation) whether an event is interpreted as a fault or a warning.**

6.2.10 Weld/No Weld

Via the output signal ”Weld/No weld” you can check whether the currently selected welding program is executed

- with current (output is set) or
- without current (output is not set).

During a welding program you can work **with current** only if

- ”Weld on External” (see page 6–5) is set and
- Weld internal (parameterizable via BOS) and
- program−related weld (parameterizable via BOS) have been switched on.

I.e. that ”Weld/No weld” is the result of an AND operation of all 3 mentioned conditions.

6.2.11 Without Monitoring

Via this output signal you can check whether the currently selected welding program is executed

- without current monitoring (output is set) or
- with current monitoring (output is not set).

CAUTION
Incorrect welding may occur!
If current monitoring is switched off, actual current values that lie outside the permissible tolerance bands do not lead to a welding fault!
Therefore, make sure that all weldings that might be executed ”without monitoring” are checked sufficiently!

During the process of a welding program, current monitoring is active only in the following cases (”Without monitoring” is not set):

- the timer−related Monitoring stopped (effective for all programs) is switched off
- the program-related current monitoring is active for **all** weld times.

6.2.12 Without Regulation and Without Monitoring

Via the ”Without regulation and without monitoring” output signal, you can check – while monitoring and regulation inhibit are deactivated for the entire module – whether all welding programs are executed with deactivated start inhibit and activated weld (firing)

- with current monitoring, and
- in CCR mode.

The output signal is not set only if exactly this is the case.
In all other situations, the output signal will be set.

CAUTION
Incorrect welding may occur!
If current monitoring is switched off, actual current values that lie outside the permissible tolerance bands do not lead to a welding fault!
Therefore, make sure that all weldings that might be executed ”without monitoring” are checked sufficiently!

6.2.13 Automatic Program Selection Active

Mirrors the state of the input signal of the same name (see page 6–4).

6.2.14 Tipdress Prewarning x

Informs the PLC/robot that the active electrode must be tipdressed as soon as possible.
If further electrode tipdressing is still allowed at this point, the signal will be set when a parameterizable wear value is reached.
The acknowledgement ”Tips have been dressed” resets ”Tipdress prewarning”.

☞ **For more information, refer to ”Electrodes have been replaced x” on page 6–9 ff.**

6.2.15 Tipdressing Request x

Informs the PLC/robot that the relevant electrode must be tipdressed.
If further electrode tipdressing is still allowed at this point, the signal will be set when a parameterizable wear value is reached.
The acknowledgement ”Tips have been dressed” resets ”Tipdressing request x”.

☞ **For more information, refer to ”Electrodes have been replaced” on page 6–9 ff.**

6.2.16 Prewarning x

Will be set when a parameterizable wear value is reached. The output signal informs the PLC/robot that the active electrode will reach the end of stepper soon and that the electrode must therefore be replaced.

A positive edge of ”End of stepper” resets ”Prewarning x”.

☞ **For more information, refer to ”Electrodes have been replaced x” on page 6–11 ff.**

6.2.17 End of Stepper Electrode x

Informs the PLC/robot that the active electrode has reached the end of stepper.

The acknowledgement ”Tips have been replaced x” resets ”End of stepper electrode x”.

☞ **It depends on the parameterization of the timer whether or not further welds are still possible after exceeding the end of stepper (parameter ”stop at end of stepper”).**

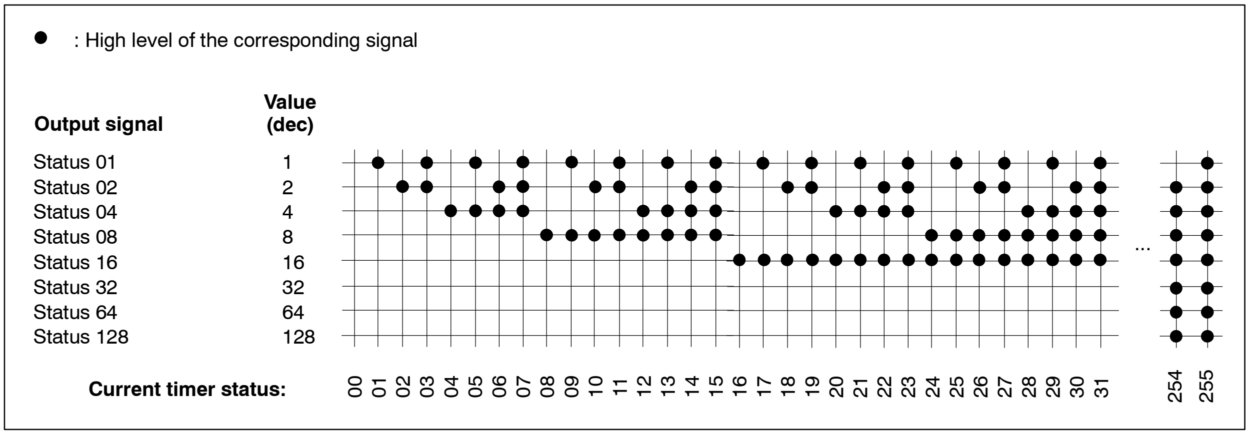
☞ **For more information, refer to ”Electrodes have been replaced x” on page 6–11.**

6.2.18 Spot Selection Taken Over

Reaction to the ”Spot selection ready” input signal.
See page 6–4 for information on the functions.

6.2.19 Status

Via the ”Status 1” through ”Status 128” outputs, the control transmits its current status in binary code via INTERBUS-S.



Binary coded output of current timer status

Currently, the following messages are defined:

Code (dec)	Description
00	OK
01	Weld off, internal
02	Weld off, external
03	Faulty program selection
04	Faulty spot selection
05	Sequence inhibited
06	No weld program
07	
08	
09	
0A	Battery error
0B	Memory cleared
0C	Hardware error
0D	External temperature too high
0E	Stop circuit open / no +24V
0F	Main switch triggered / Current passed without command
10	measuring circuit open
11	short circuit of measuring circuit
12	No primary voltage in the 1. half cycle
13	

I/O Signal Descriptions

Code (dec)	Description
14	
15	No current (standard mode)
16	No current 1. weld time (mix mode)
17	No current 2. weld time (mix mode)
18	No current 3. weld time (mix mode)
19	Current too low (standard mode)
1A	Current too low 1. weld time (mix mode)
1B	Current too low 2. weld time (mix mode)
1C	Current too low 3. weld time (mix mode)
1D	Current too high (standard mode)
1E	Current too high 1. weld time (mix mode)
1F	Current too high 2. weld time (mix mode)
20	Current too high 3. weld time (mix mode)
21	Current too low for a series of welds (standard mode)
22	Current too low for a series of welds 1. weld time (mix mode)
23	Current too low for a series of welds 2. weld time (mix mode)
24	Current too low for a series of welds 3. weld time (mix mode)
25	Time too short (standard mode)
26	Time too short 1. weld time (mix mode)
27	Time too short 2. weld time (mix mode)
28	Time too short 3. weld time (mix mode)
29	Time too long (standard mode)
2A	Time too long 1. weld time (mix mode)
2B	Time too long 2. weld time (mix mode)
2C	Time too long 3. weld time (mix mode)
2D	Minimum phase angle 1. weld time
2E	Minimum phase angle 2. weld time
2F	Minimum phase angle 3. weld time
30	Maximum phase angle 1. weld time
31	Maximum phase angle 2. weld time
32	Maximum phase angle 3. weld time
33	Full sine 1. weld time
34	Full sine 2. weld time
35	Full sine 3. weld time

I/O Signal Descriptions

Notes: